§ 121. Invariant curves of the group G_{168}.

A form of degree n,

$$\varphi(x_1, x_2, x_3), \tag{1}$$

is in general carried by the substitutions of the group into 168 different forms. For special forms φ this number can decrease; then the substitutions which leave φ unchanged form a subgroup G' contained in G_{168}, and the order of G' must divide 168. The number of forms into which φ is transformed is the index of the divisor G', and each of these different forms of φ is left fixed by a group conjugate to G'. Thus φ is an absolute invariant of G'.

The equation $\varphi = 0$ represents a curve referred to the coordinate triangle x_1, x_2, x_3. This curve is mapped by the substitutions of G_{168} to other curves. If the 168 image curves are not all distinct, then the curve φ remains fixed under the substitutions of a group G', or changes only by a constant factor; it is therefore a relative invariant of G'.

A straight line remains fixed under a substitution other than the identity only when it is one of the axes described in the preceding paragraph. All properties and relations among points, lines, and curves which are not destroyed by linear transformations, namely the projective properties of geometry, are preserved in the images. Thus if a point of a curve is a double point, an inflection point, a cusp, a point of contact of a bitangent, and so on, then its images have the corresponding property.

Among the forms φ, the important ones for us are first of all those which remain unchanged under the whole group: the invariants. Since the group is simple, there are here only absolute invariants. If $\varphi(x_1, x_2, x_3)$ is such a form, then the curve represented by $\varphi = 0$ is called an invariant curve of the group. There are then 168 mappings of the plane onto itself under which all points of the curve pass into points of the same curve. If any point lies on the curve, all its image points also lie on it.

In general the number of mutually connected points of such a curve is 168, and in any case it is not larger. If it is smaller, the point must be a pole, and the number of image points is a divisor of 168: namely 24 for the P_7, 21 for the P_8, 28 for the P_6, 42 for the P_4, 56 for the P_3, and 84 for a system of corresponding P_2. If one pole of one of these kinds lies on the curve, then all poles of the same kind lie on it.

§ 122. The first invariant of the group G_{168} and the ground curve.

To form the invariants of our group we seek forms in the three variables x_1, x_2, x_3 which remain unchanged under the three substitutions χ, r, ω. This suffices, since the whole group is generated by these three substitutions. The substitution χ cyclically permutes the variables, and r is

$$\begin{pmatrix} x_1 & x_2 & x_3 \\ \varepsilon x_1 & \varepsilon^2 x_2 & \varepsilon^4 x_3 \end{pmatrix}. \tag{2}$$

Thus, if a term $x_1^{h_1} x_2^{h_2} x_3^{h_3}$ occurs in an invariant, where the h_i are non-negative integers, invariance under r requires

$$h_1 + 2h_2 + 4h_3 \equiv 0 \pmod 7. \tag{1}$$

Invariance under χ requires that the two corresponding cyclic terms occur with it, unless $h_1 = h_2 = h_3$. There remains the condition of invariance under ω.

We first determine the smallest possible degree $m = h_1 + h_2 + h_3$. Condition (1) cannot be satisfied for $m < 3$. For $m = 3$, one would have $h_1 = h_2 = h_3 = 1$, but the product $x_1 x_2 x_3$ is not invariant under ω. The next case is $m = 4$. From

$$h_1 + h_2 + h_3 = 4, \qquad h_1 + 2h_2 + 4h_3 \equiv 0 \pmod 7 \tag{3}$$

one obtains, up to cyclic permutation, the exponent triples

$$(3,0,1), \qquad (0,3,1), \qquad (1,0,3). \tag{4}$$

Consequently, apart from a factor, the only form of degree four invariant under r and χ is

$$f(x_1,x_2,x_3) = x_1^3x_3 + x_2^3x_1 + x_3^3x_2. \tag{2}$$

To test the action of ω on f, put

$$\begin{aligned} x_1 &= \alpha y_1 + \beta y_2 + \gamma y_3, \\ x_2 &= \beta y_1 + \gamma y_2 + \alpha y_3, \\ x_3 &= \gamma y_1 + \alpha y_2 + \beta y_3, \end{aligned} \tag{3}$$

and suppose that

$$F(y_1,y_2,y_3) = f(x_1,x_2,x_3) \tag{4}$$

is thereby obtained. In forming the second derivatives one uses

$$\begin{aligned} &\tfrac{1}{6}f_{11} = x_1x_3, \quad &\tfrac{1}{6}f_{22} = x_1x_2, \quad &\tfrac{1}{6}f_{33} = x_2x_3, \\ &\tfrac{1}{3}f_{23} = x_3^2, \quad &\tfrac{1}{3}f_{31} = x_1^2, \quad &\tfrac{1}{3}f_{12} = x_2^2. \end{aligned} \tag{5}$$

The equations obtained by solving (3) have the same cyclic form. By the relations among α,β,γ from § 115 the extra coefficients vanish, and one obtains exactly the corresponding derivatives of $f(y_1,y_2,y_3)$. Euler's theorem then gives

$$F(y_1,y_2,y_3) = y_1^3y_3 + y_2^3y_1 + y_3^3y_2. \tag{6}$$

Thus f is indeed an invariant of our group G_{168}. Apart from a constant factor, it is the unique invariant of degree four.

The equation

$$f(x_1,x_2,x_3) = 0 \tag{5}$$

defines a curve of order four. We call it the ground curve of the group, or simply the curve f. There are 168 mappings of the plane onto itself under which each point of this curve corresponds to a point of the curve.

The line $x_1 = 0$ cuts the curve in three coincident points and in a fourth, separate point. The vertices of the coordinate triangle are therefore inflection points of the curve, and the sides are the inflection tangents. If the sides of the coordinate triangle are denoted by $1,2,3$ and the opposite vertices by the same numerals, then side 1 is the inflection tangent at point 2, side 2 at point 3, and side 3 at point 1.

We now examine the position of the poles and axes with respect to the ground curve. Since the vertices of the coordinate triangle are among the sevenfold poles, all P_7 lie on the ground curve. Like the vertices themselves, they are inflection points of the curve and are arranged in eight inflection triangles.

The threefold poles P_3 also lie on the ground curve. If ρ is a cube root of unity and

$$x_2 = \rho x_1, \qquad x_3 = \rho^2 x_1, \qquad 1+\rho+\rho^2 = 0, \tag{7}$$

then the point satisfies the equation of the ground curve. The joining line of the corresponding pair is

$$T = x_1 + x_2 + x_3 = 0. \tag{8}$$

Putting $x_1 = -x_2 - x_3$ in (2), the equation $f = 0$ becomes

$$(x_2^2 + x_2x_3 + x_3^2)^2 = 0. \tag{9}$$

Thus T is a bitangent. The 28 lines T are bitangents of the ground curve; their points of contact are the 56 poles P_3. The poles P_7 cannot lie on these lines.

For the points P_6, non-membership in the ground curve is immediate by putting $x_1 = x_2 = x_3 = 1$, which gives $f = 3$. For the P_8, the representation of § 118 together with the relations of § 115 likewise gives a non-zero value of f. The intersections of the 21 principal axes with one another are the poles P_8 and P_6, so two principal axes never meet on the ground curve. The intersections of a principal axis with the ground curve can therefore be only fourfold or twofold poles. If a P_4 lay both on a principal axis and on f, the axis would have to be a bitangent; but the 28 bitangents are already the lines T. This is impossible. Hence the intersections of the ground curve with the principal axes are twofold poles.

The distinguished point systems are therefore

$$24P_7,\quad 56P_3,\quad 84P_2\quad \text{on the curve } f,\qquad 21P_8,\quad 42P_4,\quad 28P_6\quad \text{not on the curve } f. \tag{10}$$

All other points of the curve f pass under the substitutions of the group into 168 different points. It follows also that the curve f has no double point and cannot decompose into curves of lower degree. For if it had a double point, all its image points would be double points; a quartic curve, unless it has a doubly counted component, cannot have more than six double points. In the exceptional case f would be the square of a quadratic form, which is impossible because there are no quadratic invariants. A system of points each of which can be transformed into any other by substitutions of the group will be called a system of connected points.

§ 123. The higher invariants.

Further invariants of our group are obtained, according to the theorem of § 40, by forming covariants of the form f. We first form the Hessian covariant

$$\Delta = \frac{1}{54}\begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix}. \tag{11}$$

For (2), up to a constant factor, this gives

$$\Delta = x_1^5 x_2 + x_2^5 x_3 + x_3^5 x_1 - 5x_1^2 x_2^2 x_3^2. \tag{3}$$

The vertices of the coordinate triangle lie on $\Delta = 0$; hence all sevenfold poles P_7 lie on Δ, and they form the complete intersection system of f and Δ.

Another covariant, now of degree fourteen, is obtained from the determinant

$$C = \frac{1}{9}\begin{vmatrix} f_{11} & f_{12} & f_{13} & \Delta_1 \\ f_{21} & f_{22} & f_{23} & \Delta_2 \\ f_{31} & f_{32} & f_{33} & \Delta_3 \\ \Delta_1 & \Delta_2 & \Delta_3 & 0 \end{vmatrix}. \tag{12}$$

It can be computed from (3), and in the present normalization is

$$\begin{aligned} C = {} & x_1^{14} + x_2^{14} + x_3^{14} \\ & - 34(x_1^{11} x_2 x_3^2 + x_1^2 x_2^{11} x_3 + x_1 x_2^2 x_3^{11}) \\ & - 250(x_1^9 x_2^4 x_3 + x_1^4 x_2 x_3^9 + x_1 x_2^9 x_3^4) \\ & + 375(x_1^8 x_2^2 x_3^4 + x_1^4 x_2^8 x_3^2 + x_1^2 x_2^4 x_3^8) \\ & + 18(x_1^7 x_2^7 + x_2^7 x_3^7 + x_3^7 x_1^7) \\ & - 126(x_1^6 x_2^5 x_3^3 + x_1^5 x_2^3 x_3^6 + x_1^3 x_2^6 x_3^5). \end{aligned} \tag{4}$$

Already from the first term one sees that the curve $C = 0$ does not pass through the vertices of the coordinate triangle.

A fourth invariant, of degree twenty-one, is obtained as the functional determinant of the three forms f, Δ, C:

$$K = \frac{1}{14}\begin{vmatrix} f_1 & \Delta_1 & C_1 \\ f_2 & \Delta_2 & C_2 \\ f_3 & \Delta_3 & C_3 \end{vmatrix}. \tag{5}$$

This curve also does not pass through the vertices of the coordinate triangle. The fully expanded expressions for C and K are the Gordan forms associated with the ternary biquadratic form f.

§ 124. The complete system of invariants.

We can now prove that the forms f, Δ, C, K are a complete invariant system of the group, that is, that every invariant of the group can be represented as an integral rational function of these four forms. We use Bezout's theorem: two curves of orders m and n which have more than mn points in common must have a common component; points of contact are to be counted twice or with their proper multiplicity.

Let Φ be an invariant of order m which does not contain f as a factor. Among the intersection points of the curves $\Phi = 0$ and $f = 0$, suppose that the poles P_7 occur h_1 times, the poles P_3 occur h_2 times, and the poles P_2 occur h_3 times; in addition, let there be h_4 systems of 168 connected points. Then

$$4m = 24h_1 + 56h_2 + 84h_3 + 168h_4, \tag{13}$$

and therefore

$$m = 6h_1 + 14h_2 + 21h_3 + 42h_4. \tag{1}$$

The numbers h_i are non-negative integers and cannot all vanish.

The smallest possible degree is thus $m = 6$. An invariant curve of degree six must pass through the 24 points P_7, as Δ does. A second invariant of degree six can therefore differ from Δ only by an expression divisible by f; since there is no invariant of degree two, it is proportional to Δ. The cases of degrees 12 and 18 similarly show that the corresponding invariants are rational functions of Δ and f.

The next case is $m = 14$. Then all invariants of order fourteen pass through the 56 points P_3, and these form the complete intersection system of such a curve with the ground curve. Hence every invariant of degree fourteen has the form

$$aC + bf\Delta, \tag{14}$$

and conversely every expression of this form is an invariant of degree fourteen.

The next degrees are treated in the same way. No independent invariant of degree twenty occurs. For degree 21 there is, apart from a constant factor, only one invariant. But the system of the 21 principal axes is itself an invariant of degree twenty-one; hence

> The invariant K decomposes into 21 linear factors, which, when set equal to zero, represent the principal axes of the group.

Next form an invariant of degree forty-two,

$$Q = \Delta^7 - kC^3, \tag{2}$$

where the constant k is chosen suitably. It can be chosen so that the curve $Q = 0$ cuts out any prescribed system of connected points on the ground curve. If Φ is any invariant not divisible by

f, we form, for the connected systems of intersection points lying on f, the corresponding forms $Q_1, Q_2, \ldots$, and put

$$W = \Phi - a\Delta^{h_1} C^{h_2} K^{h_3} Q_1 Q_2 \cdots . \tag{15}$$

The constant a can be chosen so that W passes through one more point of the ground curve; by Bezout's theorem it is then divisible by f. The quotient is again an invariant, but of lower degree. Complete induction gives the theorem:

> Every invariant of the group can be represented as an integral rational function of the four fundamental invariants f, Δ, C, K.

If the constant in (2) is chosen so that Q passes through one of the poles P_2, it follows further that $K^2 - k'Q$ is divisible by f. The quotient is an invariant of even degree and can therefore be represented without K by means of the fundamental invariants. Thus K^2 can be expressed rationally in terms of f, Δ, C. In the form

$$K^2 = \sum a_{\nu,\nu_1,\nu_2} f^{\nu} \Delta^{\nu_1} C^{\nu_2}, \qquad 2\nu + 3\nu_1 + 7\nu_2 = 21, \tag{3}$$

there is a linear relation with numerical coefficients. This relation permits all powers of K except the first to be eliminated from any expression. Hence finally:

> An invariant of even degree can be represented rationally by f, Δ, C; an invariant of odd degree is the product of K with an invariant of even degree.